

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457623.7843 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.061
Transit depth (Rp/Rs)^2	3.99 +/- 2.42 [%]
Orbital Inclination (inc)	85.71 +/- 1.87
Airmass coefficient 1 (a1)	0.76 +/- 0.38
Airmass coefficient 2 (a2)	0.46 +/- 0.45
Scatter in the residuals of the lightcurve fit is	46.77 %
Best Comparison Star	#1 - [471, 211]
Optimal Aperture	2.99
Optimal Annulus	9.65
Transit Duration (day)	0.038 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2 ± 0.061 vs 0.1592 (deviation 0.28σ)
Duration fit vs published	0.038 d vs 0.0946 d (deviation 1.17σ)
Mid-time O–C	7.8 min
Depth SNR	1.4
Residual scatter	46.77 %

Light curve

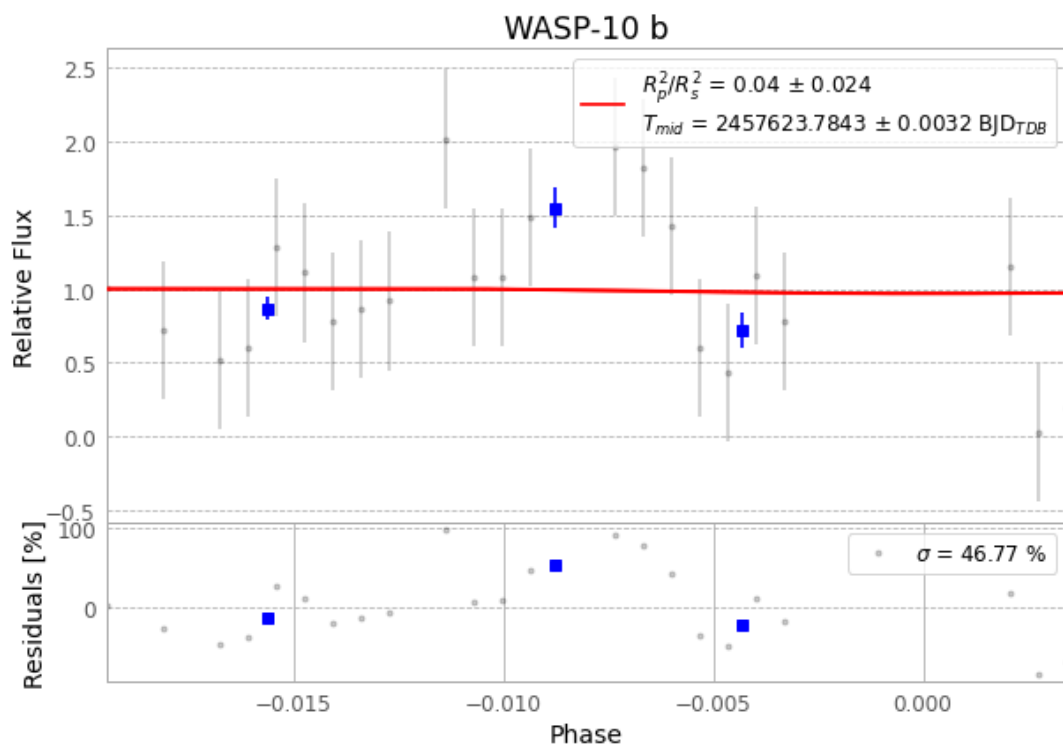


Figure 1 — FinalLightCurve_WASP-10 b_2016-08-22.png (SHA-256 8fb0e6a49be02c4a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [580, 151], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5e66061d947293cc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), WASP-10160823051512.FITS → WASP-10160823094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-160823.log (4b64641d6607...), FinalLightCurve_WASP-10 b_2016-08-22.png (8fb0e6a49be0...), FinalLightCurve_WASP-10 b_2016-08-22.csv (951663cb8cfc...)

Compute resources

Wall time 172.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:26Z.